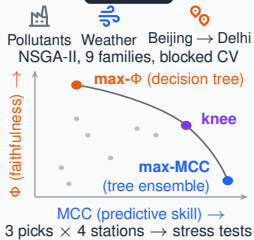


Faithfulness Is Not Robustness

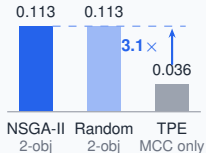
Two-objective model selection (MCC vs. SHAP faithfulness Φ) for next-day AQI-category forecasting

Setup



F1 Search

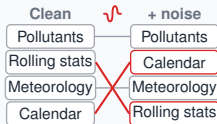
2-objective search wins;
NSGA-II itself adds little



Mean 2D hypervolume, 4 stations

F2 Simulated noise

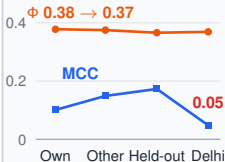
A stable Φ score hides
attribution rank collapse



Mean Φ shift ≈ 0.003 ,
yet rank ρ falls to 0.20
 Φ : tree models; ρ : worst case
Group order is schematic

F3 Transfer

For max- Φ : Φ transfers,
predictive skill does not



Delhi persistence MCC: 0.36

Takeaway: Φ alone is not a deployment signal. Report it with MCC, rank stability and a persistence baseline.